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Notes on Uncertainty in Measurements

Grado en Física

Laboratorio de Mecánica y Ondas

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Preface

The following notes are thought to provide some basic reference material for the students of “Laboratorio de Mecánica y Ondas”. Beyond presenting standard formulae and techniques, the material below intends to endow students with an essential understanding of a few fundamental concepts which underlie data and uncertainty analyses.

The adopted approach follows the international guidelines published by the Joint Committee for Guides in Metrology (JCGM), which currently comprises eight international bodies: the International Bureau of Weights and Measures (BIPM), the International Electrotechnical Commission (IEC), the International Federation of Clinical Chemistry and Laboratory Medicine (IFCC), the International Organization for Standardization (ISO), the International Union of Pure and Applied Chemistry (IUPAC), the International Union of Pure and Applied Physics (IUPAP), the International Organization of Legal Metrology (OIML), and the International Laboratory Accreditation Cooperation (ILAC). Such guidelines are collected in two main documents, the *Guide to the expression of uncertainty in measurements* (commonly known as the GUM, and updated in 2008), and the *International Vocabulary of Metrology* (VIM) (updated in 2012).¹

While being valuable, the mentioned guides will certainly appear overwhelming to an undergraduate student. Hence, a more accessible presentation of their recommendations would certainly facilitate the pursued international standard for the quantification and communication of uncertainty in measurements to be generally adopted by university laboratories. A useful overview on uncertainty analysis using the GUM is given in the book by Led Kirkup and Bob Frenkel, *An introduction to uncertainty in measurements* (Cambridge University Press, Cambridge, 2006), which can be recommended to undergraduates.²

Finally, these notes do not replace the need to consult further bibliography. A short list of selected books that students are encouraged to check, and from which some of the material contained in these notes has been collected, is included at the end and cross-referenced when appropriate throughout the text.

Salamanca, September 2025

¹The GUM and VIM are freely available at the website of the BIPM in pdf format (www.bipm.org/en/publications/guides/). Html versions are also accessible at the web site of the JCGM (www.iso.org/sites/JCGM/). A Spanish translation of the VIM is provided at the website of Centro Español de Metrología (CEM) (www.cem.es).

²This book can be downloaded from the website of Cambridge University Press at www.cambridge.org/core/books/an-introduction-to-uncertainty-in-measurement/3A67822AC6DAED5A8AC265057285AB0A from within the USAL intranet (subscription provided by the USAL library service).

Summary of key formulae

Sample mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Sample variance: $\text{var}(x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

Sample standard deviation: $s_x = \sqrt{\text{var}(x)}$

Standard uncertainty of the sample mean: $u(\bar{x}) = \frac{s_x}{\sqrt{n}}$

Degrees of freedom associated with $u(\bar{x})$: $\nu = n - 1$

Combination of $u(\bar{x})$ and limit of resolution Δ : $u(x_b) = \sqrt{u^2(\bar{x}) + \Delta^2/12}$

Degrees of freedom associated with $u(x_b)$: $\nu = (n-1) \frac{[u^2(\bar{x}) + \Delta^2/12]^2}{u^4(\bar{x})}$

Covariance between two samples: $\text{cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

Correlation coefficient: $r(x, y) = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x) \text{var}(y)}}$

Standard uncertainty of derived quantity $y = f(x_1, x_2, \dots, x_n)$ with uncorrelated inputs:

$$u(y) = \sqrt{c_1^2 u^2(x_1) + c_2^2 u^2(x_2) + \dots + c_n^2 u^2(x_n)}, \quad c_k = \frac{\partial f}{\partial x_k}$$

Weighted mean of x_j with standard uncertainties u_j :

$$\langle x \rangle = \frac{\sum_j w_j x_j}{\sum_j w_j}, \quad w_j = 1/u_j^2$$

Linear fits $y = ax + b$ [set of n pairs (x_i, y_i) of data points]:

Common uncertainty in y_i

$$a = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

$$b = \bar{y} - a \bar{x}$$

$$s_\epsilon = \sqrt{\frac{\sum_i (y_i - ax_i - b)^2}{n-2}}$$

$$u(a) = \frac{s_\epsilon}{\sqrt{(n-1) \text{var}(x)}}$$

$$u(b) = u(a) \sqrt{\langle x^2 \rangle}$$

Varying uncertainty u_i for each y_i

$$w_i = \frac{1}{u_i^2}$$

$$a = \frac{\langle xy \rangle - \langle x \rangle \langle y \rangle}{\langle x^2 \rangle - \langle x \rangle^2}$$

$$b = \langle y \rangle - a \langle x \rangle$$

$$u(a) = \frac{1}{\sqrt{\sum_i w_i} \sqrt{\langle x^2 \rangle - \langle x \rangle^2}}$$

$$u(b) = u(a) \sqrt{\langle x^2 \rangle}$$

Degrees of freedom associated with $u(a)$ and $u(b)$: $\nu = n - 2$

Effective degrees of freedom of derived quantity $y = f(x_1, x_2, \dots, x_n)$ with uncorrelated inputs:

$$\nu_{\text{eff}} = \frac{u^4(y)}{\sum_i c_i^4 u^4(x_i) / \nu_i}$$

1 Importance of measurements and their uncertainties

Experiments play a fundamental role in science, as most eloquently stated by Richard P. Feynman [1]: “*The principle of science, the definition almost, is the following: The test of all knowledge is experimental. Experiment is the sole judge of scientific ‘truth’.*”³

Experimental measurements allow us to test theories and discover new phenomena. **The result of any experimental measurement must always be a value (associated with a quantity) and its corresponding uncertainty.** The uncertainty reflects the confidence with which we believe the given value captures the ‘true’ value of the measured quantity.

What follows is an excerpt of the original paper published by Dyson, Eddington, and Davidson in 1920 (p. 328)[2], reporting the deflection of starlight by the sun’s gravitational field (measured during the total solar eclipse of May 29, 1919), which provided the first experimental evidence of Einstein’s general relativity over the Newtonian model.

As the four determinations involve only two eclipse plates and are not wholly independent, and further small accidental errors may arise through inaccurate determination of the orientation, the probable error of our mean result will be about $\pm 0''.25$. There is further the error of $\pm 0''.14$ affecting all four results equally, arising from the determination of scale. Taking this into account, and including the small correction $-0''.04$ previously mentioned, our result may be written

$$1''.61 \pm 0''.30.$$

It will be seen that the error deduced in this way from the residuals is considerably larger than at first seemed likely from the accordance of the four results. Nevertheless the accuracy seems sufficient to give a fairly trustworthy confirmation of Einstein’s theory, and to render the half deflection at least very improbable.

A nice discussion of interesting examples highlighting the importance of measurements, including the use of the laser, the discovery of new chemical elements, and the GPS system, is given in Chapter 1 of Ref. [3].

The need for ‘accurate’ measurements goes beyond purely scientific motivations. The existence of *standards of measurement*⁴ is fundamental, e.g., for international trade, and indeed for our everyday life. With the purpose of avoiding ambiguities and misunderstandings on the quantification and communication of uncertainty in measurements, in the mid 1990s, several international bodies agreed to publish the *Guide to the Expression of Uncertainty in Measurements* [4]—the GUM, which is continuously revised and updated. It must be noted, however, that, at present, the use of the GUM is not widespread in university laboratories. Here, we make an effort to adhere to the GUM recommendations.

2 Measurement fundamentals

2.1 Units and definitions

The result of a measurement is usually a number expressed as a multiple of a unit of measurement. It is therefore necessary to have an internationally agreed, comprehensive and convenient system of

³The famous Feynman Lectures on Physics are free to read online at www.feynmanlectures.caltech.edu/info/.

⁴The organization responsible for the custody, the realization, and development of the standards of measurement in a country is the corresponding national measurement institute (NMI). In Spain, this is *Centro Español de Metrología* (CEM) (www.cem.es).

Base quantity	Base unit		Factor	Prefix (Symbol)	Factor	Prefix(Symbol)
	Name	Symbol				
			10 ²⁴	yotta (Y)	10 ⁻¹	deci (d)
			10 ²¹	zetta (Z)	10 ⁻²	centi (c)
time	second	s	10 ¹⁸	exa (E)	10 ⁻³	milli (m)
length	metre	m	10 ¹⁵	peta (P)	10 ⁻⁶	micro (μ)
mass	kilogram	kg	10 ¹²	tera (T)	10 ⁻⁹	nano (n)
electric current	ampere	A	10 ⁹	giga (G)	10 ⁻¹²	pico (p)
thermodynamic temperature	kelvin	K	10 ⁶	mega (M)	10 ⁻¹⁵	femto (f)
amount of substance	mole	mol	10 ³	kilo (k)	10 ⁻¹⁸	atto (a)
luminous intensity	candela	cd	10 ²	hecto (h)	10 ⁻²¹	zepto (z)
			10 ¹	deca (da)	10 ⁻²⁴	yocto (y)

Table 1. Basic units of the SI (left) and list of SI prefixes (right). Note that the kilogram is the only SI base unit with a prefix.

units. Such is provided by the International System of Units⁵ (SI) (a brief reminder of the SI is given in Table 1).

Equally necessary is the use of an unambiguous language. In the following, we provide the definitions—along the lines of the *International Vocabulary of Metrology* (VIM) [5]— and a brief discussion of those terms that we will use most regularly.

Measurement: Process by which a value of a particular quantity is obtained.

True value: Quantity value consistent with the definition of the quantity.

The true value would be the actual value that we assume to exist and be associated with a quantity. The true value is regarded to be unknowable. It can be seen as what we would obtain for a completely specified measurand (the object of the measurement) if we could use an ideal instrument in a completely specified environment. The practical impossibility to realize such perfect conditions only allow us to get estimates of the true value.

Measurement error: Difference between the measured value and the true value of a quantity.

If the true value of a quantity were known and a measurement of that quantity yielded a deviation from the true value, then such deviation could be genuinely called a measurement error. However, as a quantity’s true value is in general unknowable, so is the measurement error.

Random measurement errors: Component of the measurement error that in replicate measurements varies in an unpredictable (random) manner.

Repeating measurements of the same quantity under what we believe to be unchanging conditions actually yield different values, due to uncontrollable fluctuations, e.g., of the measurand or of the environment. For instance,

- electronic noise when measuring a voltage with a multimeter,
- reaction times when measuring times ‘by hand’ with a stopwatch,
- fluctuations in the air temperature when measuring the speed of sound,

would be sources of random errors. The results of the measurements will thus be scattered over a range of values, as shown schematically in Fig. 1. The effect of random errors can be minimized by performing *multiple measurements* and applying *statistical techniques*.

Systematic measurement errors: Component of the measurement error that in replicate measurements remains constant or varies in a predictable manner.

⁵In November 2018, the General Conference on Weights and Measures (CGPM) made a historical decision by which from May 2019 all SI units became defined in terms of constants of nature. Visit the website of the BIPM to learn more about our current SI (www.bipm.org/en/measurement-units/).

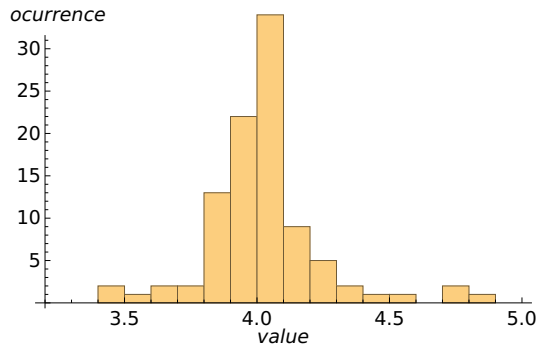


Figure 1. Measuring a quantity 100 times under the same conditions yields 100 different values, which are represented as a **histogram**. The histogram conveys the information about the frequency of occurrence of a certain value. The range of values (the x axis) is divided into constant length intervals. The height of the bars indicates the number of times the measured value falls within each interval.

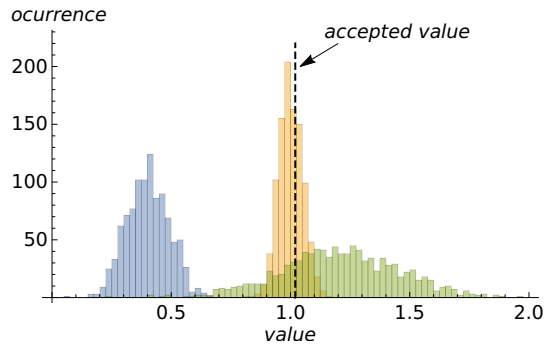


Figure 2. Histograms corresponding to three sets of 10^3 measurements of the same quantity, whose accepted value is indicated by the dashed line. The orange measurements are the most precise of the three, and also accurate. The green one may be said to be accurate but less precise. The blue one is certainly more precise than the green one, but it is not accurate, which means it is affected by systematic errors.

Systematic errors can be due, for example, to wrong calibration of a measuring device, or a minor fault such that the apparatus consistently indicates a value greater than the true value. These errors cause the measurements to shift away from the true value, and can be revealed by varying the conditions of the experiment: swapping cables, changing the measuring apparatus or the experimenter, etc. Systematic errors cannot be dealt with using statistical methods.⁶

Two common terms that are used to qualify a measurement, and that must not be confused, are **precision** and **accuracy**. Precision refers to the spread of the measurement results, whereas accuracy accounts for the closeness of the results to the true or accepted value. Thus, a very precise measurement, i.e., with a small dispersion in the observed values, may not be accurate, i.e., the measured values may not overlap with the accepted value for the quantity. This indicates that systematic errors are affecting the measurements. Conversely, a not so precise measurement may be regarded as accurate. These cases are graphically illustrated in Fig. 2.

Measurement uncertainty: Non-negative parameter characterizing the dispersion of the quantity values being attributed to a measurand.

The uncertainty associated with a measured value is therefore related to the width of the histograms or distributions observed in Figs. 1 and 2. The uncertainty indicates an interval which we believe contains the true value of the measurand to some level of confidence. **The measurement uncertainty always has a probabilistic interpretation.**⁷

The GUM classifies uncertainties in two groups. **Type A** uncertainties are those that the experimenter evaluates by statistical methods (e.g., spread of values after multiple measurements). **Type B** uncertainties are those obtained by non-statistical methods. For example, a correction in the reading given by an apparatus retrieved from a calibration report would be a type B uncertainty. The limit of resolution of a measuring device is a common type B uncertainty. The final uncertainty reported for a measured value must always include the combination of both types.

⁶One must also pay attention to another source of measurement errors insensitive to statistical treatment, namely **human mistakes** such as misreading or ignoring multiplying factors in scales, confusing units, etc.

⁷Although the GUM recommends to distinguish between the terms error and uncertainty, this is hardly done in current scientific documents. The interchangeable use of those terms does not seem to cause communication problems within the scientific community.

2.2 Scientific notation and rounding

The result of the measurement of a pendulum's period, after an appropriate uncertainty analysis, yields the value

$$T = 2.25 \pm 0.05 \text{ s.} \quad (1)$$

From the way it is written, we infer that the true value of the period lies with a high probability (we discuss how high this probability is in Secs. 3 and 6) between the values 2.20 s and 2.30 s.

When quoting the uncertainty, only a 'sensible' number of *significant figures* must be given (refresh the definition of significant figure in Sec. 2.8 of Ref. [6]). We can anticipate that in most cases the uncertainty should not have more than two significant figures.

Once the number of significant figures in the uncertainty is fixed, the quantity value must be rounded off to contain the same number of decimal places as the uncertainty. If the original value ends in 5, and this is the digit after the last one to be kept, choose the *even round value*: If the digit preceding 5 is even, leave it unchanged. If that digit is odd, increase it by one. For example, keeping two decimal digits,

3.235 is rounded to 3.24,

3.225 is rounded to 3.22.

When doing intermediate calculations keep all available digits at each step to avoid rounding errors. Once the final answer has been obtained, proceed to decide on the number of significant figures and round off the values consequently.

The use of *scientific notation* helps to avoid ambiguities when identifying significant figures: Express the quantity value with one non-zero non-decimal digit and multiply the number by the appropriate power of ten. For example,

$$1378 \text{ m} \rightarrow 1.378 \times 10^3 \text{ m,}$$

$$0.16 \text{ aC} \rightarrow 1.6 \times 10^{-19} \text{ C.}$$

Exercise 1— Consider the following table of resistance values and corresponding uncertainties. Keeping the uncertainty to two significant figures, write each resistance value using scientific notation and the $\pm$ format as in Eq. (1).

value (Ω)	uncertainty (Ω)
1000	310
100.394	1.23
99 007	278
0.123	0.0985
19.90	23.5

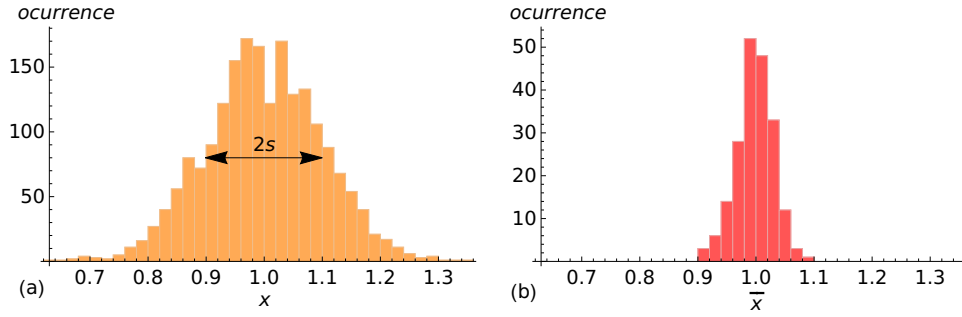


Figure 3. (a) Histogram of 2×10^3 measurements of the quantity x . The arrow marks a length of two standard deviations ($s_x = 0.101$). (b) Histogram of the 200 mean values obtained by partitioning the x_i data into 200 sets of length 10. The histogram of the means is narrower; its sample standard deviation is $u(\bar{x}) = 0.031$. According to Eq. (5), $u(\bar{x}) = s_x / \sqrt{10} = 0.032$.

3 Quantifying uncertainty: Statistical approach

3.1 Best estimate, standard uncertainty and degrees of freedom

Let $\{x_1, x_2, \dots, x_n\}$ be the set of values (a sample) obtained after n measurements of a quantity. When only random sources influence the measurement, the *best estimate* is given by the *sample mean*,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i. \quad (2)$$

When taking the mean, random errors will tend to cancel out as the number of measurements increases.

The spread of the x_i values is given by the (*unbiased*) *sample standard deviation*,

$$s_x = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}. \quad (3)$$

The square of the sample standard deviation, s_x^2 or $\text{var}(x)$, is the *sample variance*,

$$\text{var}(x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (4)$$

The standard deviation quantifies the width of the histogram of x_i [see Fig. 3(a)], and it provides the *standard uncertainty*, that in metrology is usually denoted by u . Therefore, $u(x)$ is the standard uncertainty or standard deviation of a quantity x .

The standard deviation of the mean $\bar{x}$ may be intuitively expected to be less than s_x . If we calculate many times the mean using independent sets of n measurements and construct the histogram of the means, we see that this histogram is narrower than the histogram of x , as demonstrated in Fig. 3(b). If the values x_i ($i = 1, \dots, n$) are uncorrelated (this means that they are independent from one another), the standard uncertainty of the mean is

$$u(\bar{x}) = \frac{s_x}{\sqrt{n}}, \quad (5)$$

which gives the *standard uncertainty of the best estimate*. The *relative uncertainty* is obtained as

$$\frac{u(\bar{x})}{|\bar{x}|}, \quad (6)$$

and very often expressed as a percentage.

Example 1— Consider the following measurements of the thickness W of a metallic plate: $W(\text{nm}) = \{420, 460, 400, 390, 410, 460\}$. What is the standard deviation of the observed values? Give the best estimate of the thickness, and its standard and relative uncertainties.

The best estimate is given by the mean,

$$\bar{W} = \frac{1}{6} \sum_{i=1}^6 W_i = 423.333 \text{ nm.}$$

Since we have $n = 6$ measurements, the standard deviation of the values is

$$s = \sqrt{\frac{\sum_{i=1}^6 (W_i - \bar{W})^2}{5}} = 30.111 \text{ nm.}$$

The standard uncertainty of the best estimate is

$$u = \frac{s}{\sqrt{6}} = 12.293 \text{ nm.}$$

The relative uncertainty is

$$\frac{u}{\bar{W}} \times 100\% \simeq 3\%.$$

If we choose to keep two significant figures in the uncertainty, the best estimate of the thickness can be expressed as

$$\begin{aligned} W &= (4.23 \pm 0.12) \times 10^{-7} \text{ m} \\ &= 423 \pm 12 \text{ nm.} \end{aligned}$$

Scientific notation should be used if it helps to avoid ambiguities on significant figures. In this case, however, the second form using nm as units is certainly easier to grasp.

Reporting the standard uncertainty

Besides the $\pm$ notation already seen, it is also very common to quote the standard uncertainty in the following form

$$\begin{aligned} T &= 2.25(5) \text{ s,} \\ W &= 423(12) \text{ nm,} \end{aligned}$$

where the brackets contain the uncertainty's significant figures. In fact, the GUM indicates this form as the preferred one for communicating standard uncertainties, and suggests using the $\pm$ notation only for extended uncertainties (that will be presented in Sec. 6). When reporting a quantity in the $\pm$ notation, the meaning of the quoted uncertainty should be stated explicitly.

How many significant figures should be kept in the uncertainty? This is determined by the uncertainty in the uncertainty, i.e., the standard deviation of the standard deviations. It can be shown that the relative uncertainty of the standard deviation of the mean is

$$\frac{u(u(\bar{x}))}{u(\bar{x})} \simeq \frac{1}{\sqrt{2(n-1)}}. \quad (7)$$

This means that with $n = 6$ measurements, the relative uncertainty in the uncertainty is 32%, which might suggest to keep two significant figures in $u(\bar{x})$ if the first one is 1 or 2, but only one significant figure if the first one is 3 or above. One would need around 50 measurements to bring the uncertainty

in $u(\bar{x})$ below 10%. Hence, in the experiments we will encounter in the lab, when stating the standard uncertainty as a final result, no more than two significant figures should be kept.⁸

Degrees of freedom

Note that in the sample standard deviation [Eq. (3)], the denominator is $n - 1$ when having a sample of n measurements. In fact, the differences of the values from the sample mean, $x_i - \bar{x}$, are not all independent since $\bar{x}$ has been calculated from the sample. Indeed, those differences obey $\sum_{i=1}^n (x_i - \bar{x}) = 0$, i.e., there is one restriction among them. This leaves us with $n - 1$ independent differences, and hence the $n - 1$ factor in the sample standard deviation.⁹ In this case, we say that the sample standard deviation is associated with $n - 1$ *degrees of freedom*.

The *degrees of freedom*, denoted by ν , of a set of values is the number of elements minus the number of restrictions/bindings among those elements. Suppose that we have a set of n values ϵ_i which satisfy $\sum_{i=1}^n \epsilon_i = 0$, plus some other restrictions. Then, the sample variance of the ϵ_i values would be defined as

$$s_\epsilon^2 = \frac{1}{\nu} \sum_{i=1}^n \epsilon_i^2, \quad (8)$$

and one would say that the variance is associated with ν degrees of freedom. Knowing the number of degrees of freedom associated with standard uncertainties is essential to estimate confidence intervals for the true value of a quantity, as we will see in Sec. 6.

The standard uncertainty of the sample mean [Eq. (5)] calculated from n measurements is associated with $\nu = n - 1$ degrees of freedom.

3.2 Probability density function

3.2.1 Continuous random variables

Figures 1 to 3 show histograms resulting from samples containing a finite number n of measurements. In those examples, the quantity x seems to be allowed to take with a certain probability all the possible values in a certain range. In such case we say that x is a *continuous random variable*, which is determined by a *probability density function* (PDF) $p(x)$ with the following meaning: $p(x)dx$ gives the probability that the variable takes a value within the interval $[x, x + dx]$. [We may refer to $p(x)$ simply as the distribution of x .]

As n increases, the finite intervals used to build the histograms of x could be gradually made smaller, and after normalizing the height of the bars by n , the histogram would progressively approach the probability density $p(x)$ as $n \rightarrow \infty$: The PDF describes an infinite population.

By definition, any PDF must be normalized to one,

$$\int_{-\infty}^{\infty} p(x)dx = 1, \quad (9)$$

⁸If a very large number of measurements were performed, as often occurs in an automated manner in research laboratories, it would be justified to quote the uncertainty to more significant figures, taking always into account Eq. (7).

⁹The denominator $n - 1$ then provides the unbiased sample standard deviation (and correspondingly the unbiased sample variance). The use of n as denominator in Eq. (4) instead gives a sample variance that although converging to the population variance of the underlying distribution as $n \rightarrow \infty$, does so with a biased factor of $(n - 1)/n$ [7].

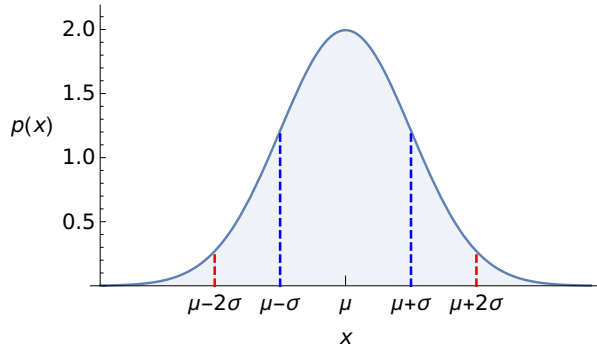


Figure 4. Gaussian distribution [Eq. (13)]. The blue (red) vertical dashed lines indicate deviations from the mean of one (two) standard deviations, and enclose 68.3% (95.4%) of the area under the curve.

where the integration runs over the real line assuming that $p(x) = 0$ in the regions not allowed for x . The probability to observe the value of x in the interval $[x_0, x_1]$ is given by

$$P_{[x_0, x_1]} = \int_{x_0}^{x_1} p(x) dx. \quad (10)$$

Important attributes of any PDF (or rather of the population it describes) are its mean, μ , and its variance, σ^2 , which are obtained as expectation values over $p(x)$,

$$\mu \equiv E(x) = \int_{-\infty}^{\infty} x p(x) dx, \quad (11)$$

$$\begin{aligned} \sigma^2 &\equiv E[(x - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx \\ &= E[x^2] - \mu^2. \end{aligned} \quad (12)$$

The square root of the variance, σ , is the standard deviation of the PDF.¹⁰

Note that the values of μ and σ can only be calculated if the distribution $p(x)$ is known. From a finite sample of values drawn from the distribution $p(x)$, the appropriate way to estimate the mean and the standard deviation of the PDF, is using $\bar{x}$ and s_x , as defined in Eqs. (2) and (3), respectively: As $n \rightarrow \infty$, $\bar{x} \rightarrow \mu$ and $s_x \rightarrow \sigma$. (Further mathematical details on this question can be found, e.g., in Refs. [7] and [8].)

In the following, some particularly relevant distributions for the understanding of measurements and their uncertainties are discussed.¹¹

3.2.2 The Gaussian distribution

The Gaussian distribution is defined as

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, \quad (13)$$

and shown in Fig. 4. This PDF is fully determined by only two parameters, μ and σ^2 , which, not by chance, correspond to its mean and to its variance, respectively.

¹⁰For completion, let us mention that the expectation value of the q -th power of x is called the q -moment of the distribution, $E(x^q)$, usually for $q \in \mathbb{N}^+$. The expectation values of $(x - \mu)^q$ are the central moments of the PDF. The mean μ is the first moment, and the variance corresponds to the second central moment.

¹¹Information on other important PDFs, such as the Poisson and the log-normal distributions can be found in Refs. [3, 6].

Exercise 2— Demonstrate that the Gaussian distribution given in Eq. (13) is normalized, and fulfils the relations

$$E(x) = \mu, \quad E[(x - \mu)^2] = \sigma^2. \quad (14)$$

Conditions that ensure the Gaussian distribution for a random variable occur rather often in nature, hence the common reference to Eq. (13) as the ‘normal’ distribution. A Gaussian random variable exhibits a characteristic symmetric distribution about the mean μ , which is the most probable value. The percentage area under the Gaussian obeys

$$\begin{aligned} [\mu - \sigma, \mu + \sigma] &\Rightarrow 68.3\%, \\ [\mu - 2\sigma, \mu + 2\sigma] &\Rightarrow 95.4\%, \\ [\mu - 3\sigma, \mu + 3\sigma] &\Rightarrow 99.7\%. \end{aligned} \quad (15)$$

Hence, these numbers indicate the probabilities for a value of x to lie in the indicated intervals.

Qualitative observation reveals that the Gaussian distribution emerges as the result of natural additive combination of independent random fluctuations. Consider a quantity influenced by various sources of random errors. For the total error to be large, all of those random independent contributions would have to act by chance in the same direction. This would happen very rarely. In contrast, such contributions are more likely to comprise, at any moment, a comparable number of positive and negative contributions, yielding a small net error. In this way, one can roughly explain the qualitative shape of the Gaussian, with a well populated central part flanked by decaying tails.

It is commonly found empirically that random errors, estimated as the deviations between measured values and their mean (or more generally a best estimate found by fitting to a model), have a Gaussian-like distribution.

Sampling from a Gaussian distribution

As mentioned earlier, if a finite sample of values is drawn from an unknown Gaussian distribution (e.g., we do several repeated measurements of a quantity that happens to obey a Gaussian PDF), the parameters μ and σ can only be estimated using the sample mean $\bar{x}$ [Eq. (2)] and the sample standard deviation s_x [Eq. (3)]. If one calculates $\bar{x}$ and s_x for many independent finite sets with n values each, histograms for $\bar{x}$ and s_x can be constructed and will have a certain width; one could thus estimate the variance of the means and the variance of the standard deviations, both of which will converge to zero as n increases, as one can intuitively expect. The standard deviation of the mean was already encountered in Eq. (5), a result which is valid for any distribution, not only for Gaussians. The variance of the sample standard deviations obeys

$$u^2(s_x) \simeq \frac{s_x^2}{2\nu}, \quad (16)$$

where ν is the number of degrees of freedom associated with s_x [cf. Eq. (7)]. The latter expression is only true for Gaussian distributions. Equation (16) can be used to assign ν to an uncertainty that is obtained when no repeated measurements are available.

As seen in Eq. (16), the number of degrees of freedom, ν , is directly related to the relative uncertainty in the uncertainty of a measurement. The larger ν the more precise the estimation of the width of the distribution is. Small ν indicates a low sampling of the distribution, and hence higher uncertainty in the estimation of its width.

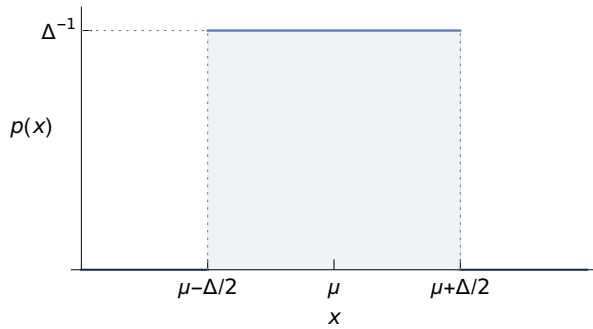


Figure 5. Uniform probability density function of width Δ [Eq. (17)].

3.2.3 The uniform distribution

The uniform distribution is defined as

$$p(x) = \begin{cases} 1/\Delta, & \mu - \Delta/2 \leq x \leq \mu + \Delta/2, \\ 0, & \text{otherwise,} \end{cases} \quad (17)$$

and is shown in Fig. 5. This PDF describes a random variable that has a constant probability of occurrence within a finite interval of values. The uniform PDF is characterized by two parameters, its mean μ and its width Δ .

Exercise 3— Demonstrate that the variance σ^2 of the uniform PDF given in Eq. (17) is

$$\sigma^2 = \frac{\Delta^2}{12}. \quad (18)$$

When all we know about a quantity are its lower and upper bounds, as, e.g., in the case of limits of resolution of measuring devices, a uniform distribution can be assumed and theoretically justified.

For example, if the voltage reading in a digital multimeter is 3.513 V—in the absence of any other information—one can only assume that the actual reading may be anywhere between 3.5125 and 3.5135 with a uniform probability.

3.3 The central limit theorem

The central limit theorem (CLT) is a fundamental theorem of Statistics. In a non-mathematical form, it can be stated as given below [3, 6, 7, 9].

Let $Z = X_1 + X_2 + \dots + X_n$ be the sum of n independent and identically distributed random variables with mean μ_X and finite variance σ_X^2 . For large n , Z will tend to be Gaussian distributed with mean $\mu_Z = n\mu_X$ and variance $\sigma_Z^2 = n\sigma_X^2$.

The direct implication of the CLT is therefore that even if X is a non-Gaussian random variable with mean μ_X and finite variance σ_X^2 , the distribution of the sample mean $\bar{X}$ obtained from n independent values of X , for large n , approaches a Gaussian of mean $\mu_{\bar{X}} = \mu_X$ and variance $\sigma_{\bar{X}}^2 = \sigma_X^2/n$.

Hence, even if a random variable is non-Gaussian, the distribution of its means will be Gaussian. When analysing the influence of random errors in measurements, the CLT is the justification to assume that the relevant PDFs are always Gaussian. The CLT is illustrated in Fig. 6.

A few additional remarks:

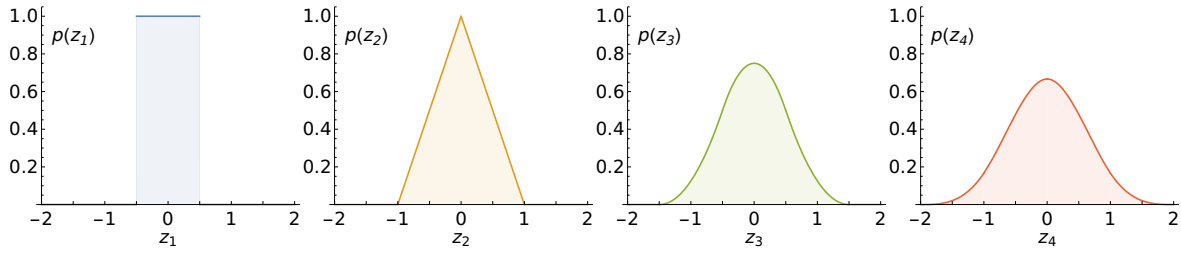


Figure 6. Probability density functions of the random variable $Z_n = X_1 + X_2 + \dots + X_n$, for $1 \leq n \leq 4$ (from left to right) where all X_i have the same uniform PDF.

- The Gaussian distribution for Z (or for $\bar{X}$) may be a good approximation already for not so large n .
- The CLT may apply even if the X_j are not identically distributed, as long as all of them contribute similarly to the sum.
- If n is large enough, the CLT may apply in some cases even if the X_j are not independent.

For completion, let us mention that the addition of any number of Gaussian random variables is always a Gaussian random variable.

The CLT helps us to understand why the Gaussian distribution appears so commonly in nature, since many observables are usually affected by multiple uncorrelated random influences. By invoking the CLT, for all practical purposes, the sample mean $\bar{x}$ obtained from a set of multiple measurements (as discussed in Sec. 3.1) will be normally distributed. Hence, we can safely apply to $\bar{x}$ all the results valid for Gaussian distributions [as we implicitly did in Eq. (7), without mentioning it].

In particular, we can now make a more precise probabilistic interpretation of the standard uncertainty of the best estimate: $\bar{x} \pm u(\bar{x})$ means that, if subsequent measurements of the mean are performed, we expect roughly 2/3 of the cases to fall in the interval $[\bar{x} - u(\bar{x}), \bar{x} + u(\bar{x})]$ [see Eq. (15)] (we elaborate further on this in Sec. 6).

4 Combining uncertainties

We now discuss the procedure to determine the best estimate and the uncertainty of a quantity that is derived from others. Let us consider that y is a function of a single quantity x ,

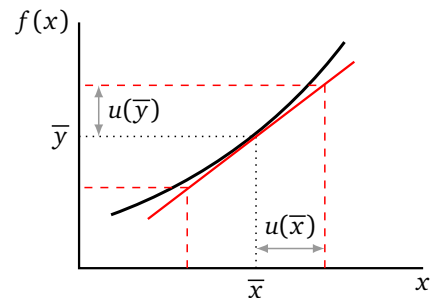
$$y = f(x). \quad (19)$$

If one had access to multiple measurements of x , one could construct all the corresponding values of y , and then calculate the mean and its corresponding uncertainty. In fact, this would be the appropriate way of doing it. However, very often only the values $\bar{x}$ and $u(\bar{x})$ are available. In that case, one relies on a *calculus approach*, under the assumption that the uncertainties are small enough.

By linearising the function $f(x)$ around the best estimate, $\bar{x}$, one can simply translate small x variations into small y variations using the first derivative. Hence,

$$\bar{y} = f(\bar{x}), \quad u(\bar{y}) = \left| \frac{df}{dx} \right|_{\bar{x}} u(\bar{x}), \quad (20)$$

as illustrated in the diagram.



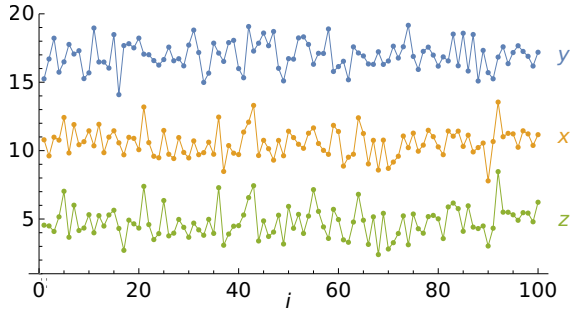


Figure 7. Correlation between 100-element samples of three quantities, x , y , and z . The correlation coefficients found according to Eq. (22) are

$$\begin{aligned} r(x, y) &= 7 \times 10^{-3}, \\ r(x, z) &= 0.9, \\ r(y, z) &= -3 \times 10^{-2}. \end{aligned}$$

When y is a function of several quantities, the combination of their uncertainties needs to be more carefully evaluated, as discussed below.

4.1 Data correlations: Covariance

Given two samples of values $\{x_i\}_{i=1,\dots,n}$ and $\{y_i\}_{i=1,\dots,n}$, of the quantities x and y , respectively, the *sample covariance* is defined as

$$\text{cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}). \quad (21)$$

Note that for the particular case $x = y$, $\text{cov}(x, x)$ corresponds to the sample variance [Eq. (4)], that we denote by s_x^2 or $\text{var}(x)$. The covariance measures the existing correlation between the samples, and would be close to zero if the samples are uncorrelated, i.e., completely independent. Correlation is often quantified using a *correlation coefficient*, defined as a suitably normalized covariance,

$$r(x, y) = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x) \text{var}(y)}}. \quad (22)$$

The correlation coefficient takes values between the limits -1 and 1 , corresponding to perfect anticorrelation and perfect correlation, respectively. Uncorrelated data will have an r close to zero. *Correlations between quantities play an important role in the propagation of uncertainties.* An example of uncorrelated and correlated samples is shown in Fig. 7.

4.2 Propagation of variances

If $z = f(x, y)$, then the **best estimate of z follows by evaluating the function at the best estimates of the inputs.**

For the sake of simplicity, here we will not use any special notation for the best estimates: From now on, the inputs x and y must be understood to be always the best estimates, with corresponding standard uncertainties $u(x)$ and $u(y)$.

The standard uncertainty of the best estimate of z obeys

$$u^2(z) = \left(\frac{\partial f}{\partial x}\right)^2 u^2(x) + \left(\frac{\partial f}{\partial y}\right)^2 u^2(y) + 2 \left(\frac{\partial f}{\partial x}\right) \left(\frac{\partial f}{\partial y}\right) \text{cov}(x, y), \quad (23)$$

where the partial derivatives are always evaluated at the inputs' best estimates. A derivation of Eq. (23) is given in Appendix A. In the particular case of uncorrelated inputs, i.e., $\text{cov}(x, y) = 0$, the

relation simplifies to

$$u^2(z) = \left(\frac{\partial f}{\partial x}\right)^2 u^2(x) + \left(\frac{\partial f}{\partial y}\right)^2 u^2(y). \quad (24)$$

This expression generalizes to more inputs, $z = f(x_1, x_2, \dots, x_n)$, as

$$u^2(z) = c_1^2 u^2(x_1) + c_2^2 u^2(x_2) + \dots + c_n^2 u^2(x_n), \quad (25)$$

where the partial derivatives are denoted by

$$c_k = \frac{\partial f}{\partial x_k}, \quad k = 1, \dots, n, \quad (26)$$

which sometimes are known as *sensitivity coefficients*.

For the case of **uncorrelated inputs**, the standard uncertainty of the best estimate of $z = f(x_1, x_2, \dots, x_n)$, is obtained by root-sum-squares of the inputs' uncertainties (either type A or type B),

$$u(z) = \sqrt{c_1^2 u^2(x_1) + c_2^2 u^2(x_2) + \dots + c_n^2 u^2(x_n)}, \quad (27)$$

where the coefficients $c_k = (\partial f / \partial x_k)$, $k = 1, \dots, n$, are evaluated at the inputs' best estimates.

Example 2— Previous measurements of the mass M and the diameter D of a steel ball bearing yield the following estimates and standard uncertainties, $M = 8.350$ g, $u(M) = 0.015$ g, $D = 12.687$ mm, $u(D) = 0.035$ mm. Determine the density of the ball bearing, and its standard and relative uncertainties.

The density reads $\rho = M/V$, where $V = \pi D^3/6$ is the volume of a sphere. In terms of M and D , one has

$$\rho = \frac{6M}{\pi D^3}. \quad (28)$$

The best estimate of ρ is obtained by evaluating this formula using the inputs' best estimates, $M = 8.350$ g and $D = 12.687$ mm, yielding

$$\rho = 7.80928 \times 10^{-3} \text{ g mm}^{-3}. \quad (29)$$

According to Eq. (27),

$$u^2(\rho) = \left(\frac{\partial \rho}{\partial M}\right)^2 u^2(M) + \left(\frac{\partial \rho}{\partial D}\right)^2 u^2(D). \quad (30)$$

The partial derivatives are

$$\frac{\partial \rho}{\partial M} = \frac{6}{\pi D^3}, \quad \frac{\partial \rho}{\partial D} = -\frac{18M}{\pi D^4}. \quad (31)$$

Therefore,

$$u^2(\rho) = \left(\frac{6}{\pi D^3}\right)^2 u^2(M) + \left(\frac{18M}{\pi D^4}\right)^2 u^2(D), \quad (32)$$

which, using the given numerical values, leads to

$$u(\rho) = 6.614 \times 10^{-5} \text{ g mm}^{-3}. \quad (33)$$

If we choose to keep the uncertainty to two significant figures, we can finally write

$$\rho = (7.809 \pm 0.066) \times 10^{-3} \text{ g mm}^{-3}. \quad (34)$$

The relative uncertainty in ρ is

$$\frac{u(\rho)}{\rho} \times 100\% = 0.85\%. \quad (35)$$

It is worth emphasizing that the composition of uncertainties often reads much simpler in terms of the relative uncertainties. For instance, the expression above for $u^2(\rho)$ can be cast as

$$\frac{u^2(\rho)}{\rho^2} = \frac{u^2(M)}{M^2} + 9 \frac{u^2(D)}{D^2}, \quad (36)$$

which eases the direct evaluation of the relative uncertainty, from which $u(\rho)$ can then be obtained.

Example 3—Equation (27) can also be applied to the case in which the inputs are n uncorrelated values of the same quantity, as it happens in the definition of the sample mean:

$$z = \frac{1}{n}(x_1 + x_2 + \dots + x_n). \quad (37)$$

Since in this case $c_k = 1/n$ for all k , the variance of z reads

$$u^2(z) = \frac{1}{n^2} [u^2(x_1) + u^2(x_2) + \dots + u^2(x_n)]. \quad (38)$$

Since all x_i are values of the same quantity, their variances are all the same and equal to the variance $s_x^2 \equiv u^2(x)$ of x . Therefore,

$$u^2(z) = \frac{s_x^2}{n} \Rightarrow u(z) = \frac{u(x)}{\sqrt{n}}. \quad (39)$$

and the usual expression for the standard uncertainty of the mean [recall Eq. (5)] is obtained.

It should be emphasized that correlations among the inputs are not always detrimental for the propagation of uncertainty (check the exercise below).

Exercise 4—Calculate the standard uncertainty of the quantity $z = x_1 - x_2$, assuming that

- a) x_1 and x_2 are perfectly correlated, i.e., $r(x_1, x_2) = 1$.
- b) x_1 and x_2 are perfectly anticorrelated, i.e., $r(x_1, x_2) = -1$.

Consider the case $u(x_1) \simeq u(x_2)$ in the final expressions.

4.3 Incorporating the limit of resolution of an instrument

Let us discuss the important issue of combining the type B uncertainty that stems from the limit of resolution of an instrument. The way in which this contribution should be taken into account is actually still a matter of research and debate (cf. page 181 in Ref. [3]). Here, we will follow the recommendations of the GUM (check Ref. [3]: Example 3 in Section 3.10, as well as the cases presented in Chapter 11.).

We will use the symbol Δ to denote the resolution of a measuring instrument. **The limits of resolution of analogue devices are taken to be the smallest readable division on the analogue scale.** For instance, the resolution of a standard ruler is usually $\Delta = 1$ mm. **For digital devices—in the absence of any additional information provided by the manufacturer—the resolution will be identified by the last decimal place indicated in the reading.** For example, the resolution of a digital multimeter that outputs a voltage reading to three decimal places would be $\Delta = 10^{-3}$ V.

Consider that the quantity x is measured multiple times in order to estimate the mean $\bar{x}$, with standard uncertainty $u(\bar{x})$. Each of the measurements is performed with a device that has a resolution

Δ . It seems logical to think that the resolution should set a lower bound for the uncertainty of the best estimate. We can think of the best estimate x_b as the combination of $\bar{x}$ and a correction C due to errors induced by the limited resolution,

$$x_b = \bar{x} + C. \quad (40)$$

The correction due to the resolution is as likely to be positive as negative, hence we take $C = 0$, i.e., $x_b = \bar{x}$. Note, however, that the uncertainty in C is not zero; it is governed by the limit of resolution. From Eq. (25), the standard uncertainty of the best estimate obeys

$$u^2(x_b) = u^2(\bar{x}) + u^2(C). \quad (41)$$

The variance $u^2(C)$ of C is determined *exactly* by assuming that the PDF associated with C is uniform with width Δ (recall the discussion on the uniform distribution in Sec. 3.2.3). In such case, $u^2(C) = \sigma^2(C)$, and we know that [recall Eq. (18)]

$$\sigma^2(C) = \frac{\Delta^2}{12}. \quad (42)$$

Therefore, the standard uncertainty of the best estimate reads

$$u(x_b) = \sqrt{u^2(\bar{x}) + \Delta^2/12}. \quad (43)$$

The minimum value of $u(x_b)$ is thus fixed by the resolution. One can also conclude that whenever the value of $u(\bar{x})$ is similar or larger than Δ , the effect of the resolution can be safely neglected.

In particular, in the case of a quantity z that is measured experimentally only once, and hence all of its uncertainty stems from the limit of resolution Δ_z , its standard uncertainty will be given by

$$u(z) = \frac{\Delta_z}{\sqrt{12}}. \quad (44)$$

The **limit of resolution Δ_z of a quantity should NOT be quoted using the $\pm$ notation**, which should be used only for standard or extended uncertainties.

Exercise 5— Revisit example 1 taking into account that the thickness measurements have a resolution of $\Delta = 10$ nm.

4.4 The weighted mean

In some situations, several independent best estimates x_j of the same quantity with different associated uncertainties $u(x_j)$ may be available. In such case, the best resulting estimate is provided by the *weighted mean*, which is denoted here by $\langle x \rangle$, and defined as

$$\langle x \rangle = \frac{\sum_j w_j x_j}{\sum_j w_j}, \quad (45)$$

where the so-called weights w_j are

$$w_j = \frac{1}{u^2(x_j)}. \quad (46)$$

The uncertainty in the resulting best estimate obeys, according to Eq. (25),

$$u^2(\langle x \rangle) = \frac{1}{\sum_j w_j} \Rightarrow \frac{1}{u^2(\langle x \rangle)} = \sum_j \frac{1}{u^2(x_j)}. \quad (47)$$

The use of the weighted mean is only sensible if it leads to a reasonably improved best estimate. In a situation where two estimates, x_1 and x_2 , are available such that $u(x_1)$ is considerably larger than $u(x_2)$, very little gain will be achieved by performing a weighted average, and one should simply use the most precise estimate available.

A situation that you might face in the laboratory is the following. Suppose one has obtained different best estimates of the same quantity, with different associated uncertainties that however make the estimates seem incompatible with one another, i.e., there is no overlapping—in the correct probabilistic sense—among the best estimates according to their uncertainties. After ruling out human mistakes, one should then conclude that there must be another unknown source of error, responsible for the apparent incompatibility of the results. In other words, the uncertainties must have been underestimated. A weighted average would be nonsensical in this context, and a sensible way of proceeding would be to discard the uncertainties and obtain a new best estimate from the mean and its corresponding standard uncertainty of the set of collected estimates.

5 Modelling data: Least-squares fits

5.1 Data visualization

Graphical representation of data is a fundamental tool in science. It is the most efficient way to report measurements, observe trends in data, look for evidence of systematic errors, estimate useful parameters, etc. The figures and plots included in a scientific report must be neat, and provide a clear, unambiguous and understandable representation of the data and the results to be communicated. To that end, a few basic guidelines are summarized below (see Chapter 5 in Ref. [6]), and a figure example shown in Fig. 8.

Guidelines for plotting data

- Tend to assign the best controlled variable (explanatory or predictor variable) to the abscissa axis (x axis), and the dependent (response) variable to the ordinate axis (y axis).
- Consider linearising the data. For instance, instead of $y = e^x$ versus x , it may be more convenient to plot $\ln y$ vs x .
- Use *appropriate scales*. For instance, the visualization of very small values may be improved if a logarithmic scale is used.
- *Label each axis* to indicate the represented variable and indicate its *units*.
- Show data points including always *error bars*. Usually the displayed error bar corresponds to $\pm$ one standard uncertainty.
- Make sure that multiple data sets are distinguishable. Use different symbols and include a legend if necessary.
- When needed, add a fit.
- Write a *proper figure caption*, where all figure elements are explained.

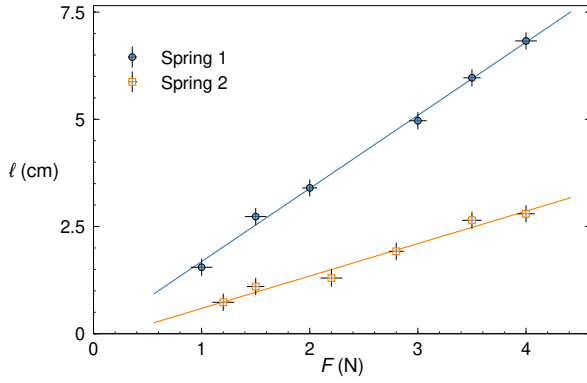


Figure 8. Experimental observation of Hooke's law. Elongation ℓ induced on two different springs by the application of a force F . Solid lines show linear fits of the form $\ell = aF + b$, considering common uncertainty in y and no error in x . The best fit parameters are $a_1 = 1.705 \pm 0.051 \text{ cm N}^{-1}$, $b_1 = -0.02 \pm 0.14 \text{ cm}$, for spring 1, and $a_2 = 0.756 \pm 0.060 \text{ cm N}^{-1}$, $b_2 = -0.17 \pm 0.16 \text{ cm}$, for spring 2. Error bars correspond to one standard deviation.

5.2 Fitting data to a model

After representing graphically a set of measurements of the quantity y for different values of the quantity x , we observe a trend in the data, or in other words, there seems to be a distinct dependence of y on x . In such case, we proceed to guess a model, i.e., a function $y = f(x; \mathbf{a})$ that may describe the data for appropriate values of the parameters $\mathbf{a} \equiv \{a_1, a_2, \dots, a_q\}$. To obtain the model parameters that provide the best fit to the data we use the *method of least squares*.

Consider the set of measurements $\{y_i\}$, with uncertainties $\{u_i \equiv u(y_i)\}$, and the corresponding values $\{x_i\}$ that are assumed to have a negligible uncertainty. We propose the model $y = f(x; \mathbf{a})$, which means

$$y_i = f(x_i; \mathbf{a}) + \epsilon_i. \quad (48)$$

measured value model value fluctuation

The differences ϵ_i between the measured and the model values are called *residuals*.

The underlying assumption to obtain the best fit is that the model value is the most probable value of the mean of the distribution of each y_i . From the CLT, one would expect the distribution of the best estimates of y_i to be Gaussian. Therefore, the probability $P(\mathbf{a})$ of occurrence of the set of values $\{y_i\}$ —provided that the values are obtained independently from one another—obeys

$$P(\mathbf{a}) \propto \exp \left[-\frac{1}{2} \sum_i \left(\frac{y_i - f(x_i; \mathbf{a})}{u_i} \right)^2 \right]. \quad (49)$$

The parameters $\mathbf{a}$ that maximize this probability would give the best fit to the data. This happens when

$$\chi^2(\mathbf{a}) \equiv \sum_i \frac{[y_i - f(x_i; \mathbf{a})]^2}{u_i^2} \quad (50)$$

is minimized.

The latter quantity is called *chi squared*, and can be seen as a weighted mean of the squares of the residuals: A data point with a large uncertainty will contribute weakly to χ^2 , whereas values with higher precision have greater importance. In the case where all the uncertainties u_i are equal, minimizing χ^2 is the same as minimizing the sum of the squares of the residuals.

Exercise 6—The sample mean is indeed the best estimate.

Let $\{x_i\}_{i=1, \dots, n}$ be a set of independently measured values of the same quantity x , where all the values are characterized by the same standard uncertainty. Do a least-squares fit of the data to a one-parameter model, i.e., $x_i = a + \epsilon_i$, and obtain the value of the parameter a .

Consider now that the uncertainty of each x_i is different, namely u_i , and find the best one-parameter fit by minimizing χ^2 .

The estimates of the best fit parameters, $\mathbf{a}$, (i.e., those that minimize χ^2) have associated uncertainties, $u(\mathbf{a})$, inherited from the existing uncertainty in the y_i values, which manifests itself in the fluctuation of the data around the best fit.

Since the set of data is used to estimate the q parameters of the fit, one can intuitively see that the residuals will be bound by q equations, i.e., q restrictions (recall the discussion on degrees of freedom in Sec. 3.1). As a consequence, the standard uncertainties of the best fit parameters are associated with $n - q$ degrees of freedom.

A fit of n data points to a model with q parameters is said to have

$$\nu = n - q \quad (51)$$

degrees of freedom. Strictly speaking this means that the uncertainties $u(\mathbf{a})$ of the best fit parameters are associated with $\nu = n - q$ degrees of freedom.

5.2.1 Linear fit of data with common uncertainties

Let us consider the case of a linear model,

$$f(x) = ax + b, \quad (52)$$

which has two parameters, the slope a and the ordinate at the origin, b . In order to find the best linear fit of the data $\{y_i, x_i\}_{i=1, \dots, n}$, assuming that all the y_i have a common uncertainty, one must minimize the sum of the squares of the residuals,

$$Q^2(a, b) = \sum_i (y_i - ax_i - b)^2. \quad (53)$$

The minimization of Q^2 over a and b yields the following best fit parameters,

$$a = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2}, \quad (54a)$$

$$b = \bar{y} - \bar{x}a = \frac{\bar{y}\overline{x^2} - \bar{x}\overline{xy}}{\overline{x^2} - \bar{x}^2}, \quad (54b)$$

where, as usual, the overline denotes the arithmetic mean. For instance,

$$\overline{x^2} = \frac{1}{n} \sum_{i=1}^n x_i^2, \quad \overline{xy} = \frac{1}{n} \sum_{i=1}^n x_i y_i. \quad (55)$$

If the linear model is an acceptable description of the data, the scatter of the residuals (i.e., the scatter of the y_i data around the best fit) must reflect the common uncertainty of the y_i values. Once a and b are known, the standard deviation s_ϵ of the residuals is given by

$$s_\epsilon = \sqrt{\frac{\sum_i (y_i - ax_i - b)^2}{n - 2}}, \quad (56)$$

where the factor $n - 2$ in the denominator corresponds to the number of degrees of freedom. This is sometimes called the standard deviation of the fit, or root-mean-square residual.

The standard uncertainties of a and b can be obtained using Eq. (27) and taking s_ϵ to be the common uncertainty of y_i . This leads to

$$u(a) = \frac{s_\epsilon}{\sqrt{(n-1)\text{var}(x)}} = \frac{s_\epsilon}{\sqrt{n(\overline{x^2} - \bar{x}^2)}}, \quad (57a)$$

$$u(b) = u(a)\sqrt{\overline{x^2}}. \quad (57b)$$

Note that these uncertainties are of order $\sqrt{n}$ less than s_ϵ . The derivation of Eqs. (54) and (57) is presented in Appendix B.

The correlation coefficient $r(x, y)$, defined in Eq. (22), gives some information about the plausibility of the linear dependence between x and y . A perfect linear correlation between the variables implies $|r| = 1$. Hence, the closer $|r|$ is to one, the stronger the evidence of a linear model.

Exercise 7— Demonstrate that if the set of values $\{y_i\}$ exhibits a perfect linear dependence on the set $\{x_i\}$, i.e., $y_i = ax_i + b$, for constant a and b , then

$$|r(x, y)| = 1. \quad (58)$$

The relative uncertainty of the best fit parameter a , can be written in terms of the correlation coefficient $r(x, y)$ as

$$\frac{u(a)}{|a|} = \sqrt{\frac{1}{n-2} \left(\frac{1}{r^2} - 1 \right)}, \quad (59)$$

as demonstrated in Appendix B.

Exercise 8— The table below lists the data plotted in Fig. 8.

	Spring 1						Spring 2					
$F(\text{N})$	1.0	1.5	2.0	3.0	3.5	4.0	1.2	1.5	2.2	2.8	3.5	4.0
$\ell(\text{cm})$	1.55	2.73	3.40	4.97	5.97	6.83	0.73	1.10	1.30	1.92	2.65	2.80

Perform linear fits, ℓ versus F , assuming common uncertainties for the elongation (ℓ) values. Compare the fit parameters and their uncertainties to the values given in the caption of Fig. 8.

5.2.2 Linear fit of data with varying uncertainties

When the values y_i have varying uncertainties u_i the best linear fit should be found by minimizing χ^2 ,

$$\chi^2(a, b) = \sum_i w_i (y_i - ax_i - b)^2, \quad (60)$$

where the weights w_i are determined by the y_i uncertainties,

$$w_i = \frac{1}{u_i^2}. \quad (61)$$

The minimization over a and b leads to expressions very similar to Eqs. (54), but with weighted averages instead of arithmetic averages,

$$a = \frac{\langle xy \rangle - \langle x \rangle \langle y \rangle}{\langle x^2 \rangle - \langle x \rangle^2}, \quad (62a)$$

$$b = \langle y \rangle - a \langle x \rangle, \quad (62b)$$

where all weighted averages are carried out using the weights w_i . For instance,

$$\langle x^2 \rangle = \frac{\sum_{i=1}^n w_i x_i^2}{\sum_i w_i}, \quad \langle xy \rangle = \frac{\sum_{i=1}^n w_i x_i y_i}{\sum_i w_i}. \quad (63)$$

The application of Eq. (27) to a and b yields the parameter uncertainties

$$u(a) = \frac{1}{\sqrt{\sum_i w_i} \sqrt{\langle x^2 \rangle - \langle x \rangle^2}}, \quad (64a)$$

$$u(b) = u(a) \sqrt{\langle x^2 \rangle}. \quad (64b)$$

We have not considered the case in which the values x_i have non-negligible uncertainties. Finding the best linear fit in that case can be formulated in terms of the minimization of a modified χ^2 (see the brief discussion in Chapter 9 of Ref. [6]), but it becomes a much harder problem involving non-linear equations for the parameters.

A variety of model fits are nowadays implemented in spreadsheets and mathematical software packages. When using those, one must carefully read the available documentation to understand the data processing that is performed.

5.2.3 Is the model acceptable?

One can always attempt a fit of any set of data to a certain model, but that does not make the model acceptable. Checking the validity of the model hypothesis is an interesting topic which requires further concepts beyond the scope of these notes. Useful information can be found in Ref. [6]. We will mention, however, a couple of helpful pointers.

Regarding the linear model, we already discussed that a qualitative indication of its plausibility is provided by the correlation coefficient $r(x, y)$. Most importantly,

- when Q^2 is minimized, the fit standard deviation s_e [Eq. (56)] should be similar to the common uncertainty of the data y_i (assuming such uncertainty to be known).
- when χ^2 is minimized, the minimum chi square should be comparable to the number of degrees of freedom, e.g., $\chi_{\min}^2 \approx n - 2$ for a linear fit.

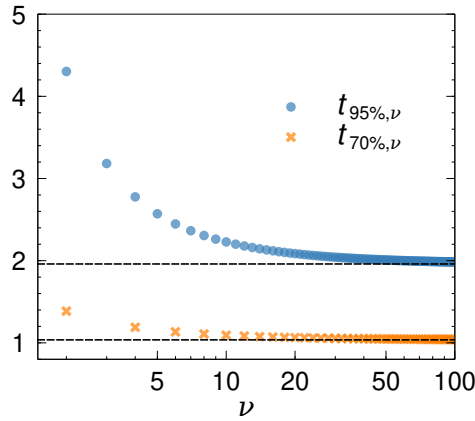
An exploration of the residuals of the best fit also gives valuable information about the reliability of the proposed model (e.g., they should be nearly normally distributed, and should not exhibit any distinctive trends) [6].

6 Coverage intervals (Extended uncertainty)

6.1 Coverage factors

At this point we know how to calculate the standard uncertainty of a quantity's best estimate. Most importantly, we know how to interpret it: By invoking the CLT and using the properties of the Gaussian distribution, we know that the standard uncertainty captures the true value with a probability of roughly 2/3.

Very often, it is necessary or desirable to provide an *extended uncertainty* (also called *coverage interval* or *confidence interval*) within which the true value of the measurand is located with higher probability, usually 95%. From the properties of the normal distribution, this interval corresponds



ν	$t_{95\%, \nu}$	ν	$t_{95\%, \nu}$
2	4.30	9	2.26
3	3.18	10	2.23
4	2.78	15	2.13
5	2.57	20	2.09
6	2.45	50	2.01
7	2.36	100	1.98
8	2.31	∞	1.96

Figure 9. Evolution of the 95% coverage factors $t_{95\%, \nu}$ (blue circles) as a function of the number of degrees of freedom ν . For comparison, the 70% coverage factors $t_{70\%, \nu}$ (orange crosses) are also shown in the plot.

to two standard deviations (more accurately to $\pm 1.96\sigma$) from the mean μ [see Eqs. (15)]. However, we only have access to an estimate of the true mean μ and of the true σ via the sample mean $\bar{x}$ and the sample standard deviation s_x , both of which will change with the number of measurements. In particular, recall that there is an uncertainty in the sample standard deviation given by Eq. (16).

As you may guess, the precise multiplicative factor of the standard uncertainty in order to attain the extended uncertainty depends on the number of measurements considered in our estimation of s_x , more precisely, it depends on the number of degrees of freedom ν associated with s_x . Recall, that ν is actually what determines the uncertainty in the uncertainty.

If we denote the extended uncertainty of the best estimate by $U(x)$, then

$$U(x) = t_{95\%, \nu} u(x), \quad (65)$$

where $u(x)$ denotes the standard uncertainty of the best estimate, ν is the number of degrees of freedom associated with $u(x)$, and $t_{95\%, \nu}$ is the coverage factor or Student t value.¹² The dependence of the 95% coverage factors on ν is shown in Fig. 9.

Therefore, if $u(x)$ is associated, for example, with $\nu = 6$ degrees of freedom, then $U(x) = 2.45 u(x)$. **Extended uncertainties are usually quoted to two significant figures.**

According to the GUM, the preferred notation to report a quantity with its extended uncertainty should be the $\pm$ notation, i.e., $x \pm U(x)$. Nonetheless, it is rather common in the scientific literature to find 95% confidence intervals reported as numerical intervals, i.e., $[x - U(x), x + U(x)]$. (Example: Best estimate $x = 2.25$ with extended uncertainty $U(x) = 0.13$, then the 95% confidence interval would be quoted as $[2.12, 2.38]$.) Independently of the notation chosen, one should always state explicitly the meaning of the quoted uncertainty in order to avoid misunderstandings.

6.2 The Welch-Satterthwaite formula

Consider the derived quantity $y = f(x_1, x_2, \dots, x_n)$. As we saw in Sec. 4, assuming uncorrelated input variables x_i , the variance of y obeys

$$u^2(y) = c_1^2 u^2(x_1) + \dots + c_n^2 u_n^2(x_n), \quad (66)$$

¹²Named after the pseudonym ‘Student’ used by W. S. Gosset, who studied this problem and published his findings in 1906.

in terms of the standard uncertainties of the inputs' best estimates and the sensitivity coefficients c_i [Eq. (26)]. If each uncertainty $u(x_i)$ is associated with ν_i degrees of freedom, how many degrees of freedom should one associate with $u(y)$?

The latter question can be answered by calculating the variance of the variances in the equation above and using Eq. (16). The resulting expression for the effective number of degrees of freedom associated with $u(y)$ is given by the *Welch-Satterthwaite formula*,¹³

$$\nu_{\text{eff}} = \frac{u^4(y)}{\sum_i c_i^4 \frac{u^4(x_i)}{\nu_i}}, \quad (67)$$

where the result is truncated to the closest integer from below. From the value of ν_{eff} , one can now determine the extended uncertainty for y .

It can also be shown that

$$\min(\nu_i) \leq \nu_{\text{eff}} \leq \sum_i \nu_i. \quad (68)$$

Let us consider the simple case of two inputs such that $c_1 = c_2 = 1$, then

$$\nu_{\text{eff}} = \frac{[u^2(x_1) + u^2(x_2)]^2}{\frac{u^4(x_1)}{\nu_1} + \frac{u^4(x_2)}{\nu_2}}. \quad (69)$$

If, additionally, their uncertainties are similar,

$$\nu_{\text{eff}} = \frac{4\nu_1\nu_2}{\nu_1 + \nu_2}. \quad (70)$$

It is worth discussing the special case of the uncertainty stemming from the limit of resolution of an instrument. As presented in Sec. 4.3, in this case one claims to know the (uniform) distribution exactly, and thus to have its standard deviation with no uncertainty. This therefore corresponds to having an infinite number of degrees of freedom. Consequently, **a quantity whose standard uncertainty has been estimated only from an instrument's limit of resolution does not contribute to the denominator of the Welch-Satterthwaite formula.**

Consider the situation where the limit of resolution Δ is combined with the mean $\bar{x}$ to obtain a best estimate, as done in Sec. 4.3. If the number of degrees of freedom associated with $u(\bar{x})$ is $n-1$, then ν_{eff} after including the limit of resolution, is

$$\nu_{\text{eff}} = (n-1) \frac{[u^2(\bar{x}) + \Delta^2/12]^2}{u^4(\bar{x})}. \quad (71)$$

When using the Welch-Satterthwaite formula, due to the fourth powers there involved, it is advisable to use all available decimal digits when evaluating standard uncertainties, avoiding round-off errors for the effective number of degrees of freedom and for the coverage intervals.

Example 4— The moment of inertia, I , of a solid cylinder of mass M about its symmetry axis is

¹³In the derivation on Eq. (67) one needs to assume that not only the inputs $x_1, \dots, x_n$ but also their variances are uncorrelated.

given by

$$I = \frac{1}{2}MR^2, \quad (72)$$

where R is the radius of the cylinder. After eight measurements, the best estimate of the mass is determined to be $M = 252.6 \text{ g}$ with standard uncertainty $u(M) = 2.5 \text{ g}$. The radius is measured five times, yielding the best estimate $R = 6.35 \text{ cm}$ with standard uncertainty $u(R) = 0.05 \text{ cm}$. Assuming that the errors in the measurements of the mass and the radius are uncorrelated, determine

- (a) the best estimate of the moment of inertia and its standard uncertainty, $u(I)$,
- (b) the effective number of degrees of freedom associated with $u(I)$,
- (c) the 95% confidence interval for the true value of I .

From the numerical values of M and R , the best estimate of the moment of inertia is

$$I = 5092.73 \text{ g cm}^2. \quad (73)$$

The sensitivity coefficients are

$$c_M = \partial_M I = \frac{R^2}{2}, \quad c_R = \partial_R I = MR. \quad (74)$$

Therefore

$$\begin{aligned} u^2(I) &= \frac{R^4}{4}u^2(M) + M^2R^2u^2(R) \\ &= 8972.6 \text{ g}^2\text{cm}^4, \end{aligned} \quad (75)$$

and $u(I) = 94.72 \text{ g cm}^2$. Keeping $u(I)$ to two significant figures, the best estimate of I with its standard uncertainty could be reported as

$$I = 5.093(95) \text{ kg cm}^2. \quad (76)$$

The effective number of degrees of freedom associated with $u(I)$ is given by the Welch-Satterthwaite formula,

$$\nu_{\text{eff}} = \frac{u^4(I)}{c_M^4 \frac{u^4(M)}{\nu_M} + c_R^4 \frac{u^4(R)}{\nu_R}}. \quad (77)$$

Since $\nu_M = 7$ and $\nu_R = 4$, numerical substitution leads to

$$\nu_{\text{eff}} = 7, \quad (78)$$

after truncating to the closest integer from below. The 95% coverage factor for 7 degrees of freedom is 2.36. The extended uncertainty is therefore $U(I) = 2.36u(I) = 223.5 \text{ g cm}^2$. Finally, the best estimate of I with its corresponding extended uncertainty is

$$I = 5.09 \pm 0.22 \text{ kg cm}^2. \quad (79)$$

Exercise 9— The diameter D of a wire is measured five times using a micrometer with a resolution of 0.01 mm. The mean diameter is found to be 0.253 mm with a standard uncertainty of 0.007 mm. Calculate

- (a) the best estimate of the cross-sectional area of the wire, $A = \pi D^2/4$, and its standard

uncertainty,

(b) the effective number of degrees of freedom associated with $u(A)$,

(c) the 95% confidence interval for the true value of A .

Chapter 11 of Ref. [3] is entirely devoted to show case studies and examples based on undergraduate experiments, where the calculation of standard and extended uncertainties is presented in detail.

7 Elaborating a scientific report

Here, we give some helpful indications on how to elaborate the reports of the experiments performed in the lab. Most of these comments are indeed relevant for the preparation of any scientific document.

- A scientific report should be clear and useful to others. In particular, all the steps followed in measurements and in subsequent data analyses must be concisely and unambiguously described for others to be able to reproduce and verify your results.
- For the sake of clarity, the report should be organized in different sections. The authors have certainly the freedom to choose the optimal distribution of the contents. Nonetheless, a sensible sectioning would usually not deviate much from the following.
 1. **Introduction/Theoretical background.** Description of the theory behind the experimental phenomenology studied, along, perhaps, with motivations and goals of the experiment.
 2. **Experimental procedure/Measurements.** Description of the procedure followed to perform measurements, and collection of measured data.
 3. **Data analysis and results.** Further analysis of data, calculation and presentation of results.
 4. **Final discussion/Conclusions.** Discussion on the meaning and implication of results, difficulties found, whether inconsistencies have been observed and what could explain them, things that could be improved, etc.
- When reporting direct measurements, organize values in tables indicating clearly the corresponding limits of resolution and/or standard uncertainties. **All directly measured quantities must be included in a scientific report.**
- Use symbols appropriately and consistently to denote physical quantities.
- Use common sense when quantifying limits of resolution. The conditions under which a measurement is performed may set a larger limit of resolution than the one of the device used to measure.
- When doing data analysis, report values of intermediate calculations organized in tables (when appropriate) indicating the corresponding standard uncertainties. In particular, report always the data used in linear fits.
- When performing linear fits, report the values of all fit parameters and their standard uncertainties.
- Include neat and useful plots (recall the guidelines given in Sec. 5.1).

- Keep all digits in intermediate calculations to avoid accumulating rounding errors. Rounding should only be done at the very end when reporting final results.
- When stating final results, do not quote standard and extended uncertainties to more than two significant figures.
- Tables and figures should be included as labelled floating objects which are cross-referenced in the text.
- Try to make a consistent use of the typefaces used for symbols, in line with the accepted international guidelines described, e.g., in physics.nist.gov/cuu/pdf/typefaces.pdf.
- Revise—very carefully—the text and the contents of the report before handing it in. Understand that all authors of a scientific document are responsible for the contents and claims there contained.

In closing, let us mention that, although here we made an effort to follow the recommendations of the GUM for the calculation and communication of uncertainty in measurements, it is very common to find in current scientific literature that different authors apply slightly different approaches to this problem. It is in fact very likely, that you, as an undergraduate student, will be told to do the uncertainty analysis in different ways across different labs in your degree. Therefore, when writing a scientific report, it is of fundamental importance to explain how the uncertainty analysis was carried out, thereby allowing other colleagues, even if they disagree on the method, to correctly interpret and understand your results.

A Propagation of variances

Let us consider the quantity $z = f(x, y)$, having a functional dependence on inputs x and y , which are always understood as best estimates. Assume that the result of a measurement of the inputs yields the values x_i and y_i . Correspondingly, one obtains the value of z ,

$$z_i = f(x_i, y_i). \quad (80)$$

Multiple measurements of the best estimates would allow to calculate the means of the best estimates, denoted by $\bar{x}$ and $\bar{y}$ (these can be seen as the means of the means). A Taylor expansion of z_i around the means to first order gives

$$z_i \simeq f(\bar{x}, \bar{y}) + \left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} (x_i - \bar{x}) + \left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} (y_i - \bar{y}). \quad (81)$$

By taking into account all multiple measurements, $\bar{z}$ can be calculated,

$$\sum_{i=1}^n z_i \simeq n f(\bar{x}, \bar{y}) + \left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} \underbrace{\sum_{i=1}^n (x_i - \bar{x})}_0 + \left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} \underbrace{\sum_{i=1}^n (y_i - \bar{y})}_0. \quad (82)$$

Hence

$$\bar{z} \simeq f(\bar{x}, \bar{y}). \quad (83)$$

It is important to note, however, that higher order terms in the above Taylor expansion would yield non-vanishing contributions to $\bar{z}$. If the function f is highly nonlinear, further corrections to $\bar{z}$ may be necessary.

Let us now proceed to estimate the uncertainty in z . From Eqs. (81) and (83), one has

$$\begin{aligned} (z_i - \bar{z})^2 &\simeq \left[\left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} (x_i - \bar{x}) + \left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} (y_i - \bar{y}) \right]^2 \\ &\simeq \left(\left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} \right)^2 (x_i - \bar{x})^2 + \left(\left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} \right)^2 (y_i - \bar{y})^2 + 2 \left(\left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} \right) \left(\left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} \right) (x_i - \bar{x})(y_i - \bar{y}). \end{aligned} \quad (84)$$

By summing over all multiple measurements, and recalling the definitions of the variance [Eq. (4)] and the covariance [Eq. (21)], one finally gets to

$$\text{var}(z) = \left(\left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} \right)^2 \text{var}(x) + \left(\left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} \right)^2 \text{var}(y) + 2 \left(\left. \frac{\partial f}{\partial x} \right|_{\bar{x}, \bar{y}} \right) \left(\left. \frac{\partial f}{\partial y} \right|_{\bar{x}, \bar{y}} \right) \text{cov}(x, y), \quad (85)$$

where the partial derivatives are evaluated at the inputs' best estimates, and we recall that the variances and covariances refer to those of and between best estimates.

The latter expression is easily generalized to the case of multiple inputs, $z = f(x_1, x_2, \dots, x_n)$,

$$\text{var}(z) = \sum_{i=1}^n \sum_{j=1}^n \left(\left. \frac{\partial f}{\partial x_i} \right|_{\bar{x}_i} \right) \left(\left. \frac{\partial f}{\partial x_j} \right|_{\bar{x}_j} \right) \text{cov}(x_i, x_j), \quad (86)$$

where $\text{cov}(x_i, x_i) = \text{var}(x_i)$. If all input quantities are independent, i.e., $\text{cov}(x_i, x_j) = 0$ for $i \neq j$, the expression simplifies to

$$\text{var}(z) = \sum_{i=1}^n \left(\left. \frac{\partial f}{\partial x_i} \right|_{\bar{x}_i} \right)^2 \text{var}(x_i). \quad (87)$$

B Least-squares linear fit

The best linear fit of the data $\{y_i, x_i\}_{i=1,\dots,n}$, assuming that y_i have common uncertainty, follows from the minimization of the sum of the squares of the residuals,

$$Q^2(a, b) = \sum_i \epsilon_i^2, \quad (88)$$

where

$$\epsilon_i = (y_i - ax_i - b). \quad (89)$$

At the Q^2 minimum, the partial derivatives with respect to a and b must vanish:

$$\partial_a Q^2 = -2 \sum_i \epsilon_i x_i = 0, \quad (90a)$$

$$\partial_b Q^2 = -2 \sum_i \epsilon_i = 0. \quad (90b)$$

We see that the minimization condition translates into two restrictions among the residuals, i.e., the best fit residuals have $n - 2$ degrees of freedom. Additionally, to check that the solution to Eqs. (90) is a minimum, one needs to construct the Hessian matrix H (the matrix of second derivatives):

$$\begin{aligned} \partial_a^2 Q^2 &= 2n \overline{x^2}, \\ \partial_b^2 Q^2 &= 2n, \\ \partial_a \partial_b Q^2 &= 2n \bar{x} \end{aligned} \Rightarrow H = 2n \begin{pmatrix} \overline{x^2} & \bar{x} \\ \bar{x} & 1 \end{pmatrix}. \quad (91)$$

Since $\det H = 2n(\overline{x^2} - \bar{x}^2) = 2(n-1)\text{var}(x) > 0$ and $\partial_a^2 Q^2 > 0$, the critical point is a minimum.

From Eqs. (90), one has

$$\overline{xy} - a \overline{x^2} - b \bar{x} = 0, \quad (92)$$

$$\bar{y} - a \bar{x} - b = 0, \quad (93)$$

which can be straightforwardly solved to get

$$a = \frac{\overline{xy} - \bar{x} \bar{y}}{\overline{x^2} - \bar{x}^2} = \frac{\text{cov}(x, y)}{\text{var}(x)}, \quad (94)$$

$$b = \bar{y} - a \bar{x}. \quad (95)$$

Note that the sample variance and covariance, defined in Eqs. (4) and (21), respectively, can be written as

$$\text{var}(x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n}{n-1} (\overline{x^2} - \bar{x}^2), \quad (96)$$

$$\text{cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{n}{n-1} (\overline{xy} - \bar{x} \bar{y}). \quad (97)$$

The scatter of the y_i data around the best fit is characterized by the standard deviation of the fit, or the root-mean-square residual,

$$s_\epsilon = \sqrt{\frac{\sum_i (y_i - ax_i - b)^2}{n-2}}. \quad (98)$$

The uncertainty of the best fit parameters is obtained using Eq. (25), taking s_ϵ as the uncertainty of each y_i value. Since

$$\begin{aligned}\frac{\partial \text{cov}(x, y)}{\partial y_j} &= \frac{n}{n-1} \frac{\partial}{\partial y_j} \left(\frac{1}{n} \sum_i x_i y_i - \bar{x} \frac{1}{n} \sum_i y_i \right) \\ &= \frac{n}{n-1} \left(\frac{1}{n} x_j - \frac{1}{n} \bar{x} \right) \\ &= \frac{1}{n-1} (x_j - \bar{x}),\end{aligned}\quad (99)$$

it ensues

$$\begin{aligned}u^2(a) &= \frac{1}{\text{var}^2(x)} \sum_j \frac{1}{(n-1)^2} (x_j - \bar{x})^2 s_\epsilon^2 \\ &= \frac{s_\epsilon^2}{(n-1) \text{var}(x)} = \frac{s_\epsilon^2}{n(\overline{x^2} - \bar{x}^2)}.\end{aligned}\quad (100)$$

And from the definition of b ,

$$\begin{aligned}u^2(b) &= u^2(\bar{y}) + \bar{x}^2 u^2(a) \\ &= \frac{s_\epsilon^2}{n} + \bar{x}^2 \frac{s_\epsilon^2}{n(\overline{x^2} - \bar{x}^2)} \\ &= \frac{s_\epsilon^2}{n(\overline{x^2} - \bar{x}^2)} \bar{x}^2 = u^2(a) \bar{x}^2.\end{aligned}\quad (101)$$

The correlation coefficient $r \equiv r(x, y)$ can be written as

$$r = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x) \text{var}(y)}} = a \sqrt{\frac{\text{var}(x)}{\text{var}(y)}}.\quad (102)$$

Since $\sum_i \epsilon_i = 0$ [see Eqs. (90)], one has $y_i - \bar{y} = a(x_i - \bar{x}) - \epsilon_i$, and $\text{var}(y)$ can be expressed as

$$\begin{aligned}\text{var}(y) &= \frac{1}{n-1} \sum_i [a^2(x_i - \bar{x})^2 + 2\epsilon_i(x_i - \bar{x}) + \epsilon_i^2] \\ &= a^2 \text{var}(x) + \frac{1}{n-1} \sum_i \epsilon_i^2 = a^2 \text{var}(x) + \frac{n-2}{n-1} s_\epsilon^2,\end{aligned}\quad (103)$$

using the restrictions of the residuals given in Eqs. (90), and the definition of s_ϵ . Therefore,

$$\begin{aligned}r &= \frac{a \text{var}(x)}{\sqrt{\text{var}(x) [a^2 \text{var}(x) + \frac{n-2}{n-1} s_\epsilon^2]}} \\ &= \left[1 + \frac{n-2}{a^2} \frac{s_\epsilon^2}{(n-1) \text{var}(x)} \right]^{-1/2} \\ &= \left[1 + (n-2) \frac{u^2(a)}{a^2} \right]^{-1/2}.\end{aligned}\quad (104)$$

Or equivalently,

$$\frac{u^2(a)}{a^2} = \frac{1}{n-2} \left(\frac{1}{r^2} - 1 \right).\quad (105)$$

For the minimization of χ^2 , the same procedure presented here can be followed without further difficulty to obtain the best fit parameters and their standard uncertainties.

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